



MEC6602E – Transsonic Aerodynamics

Homework 1 : 1D Wave and Euler Equations

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1 One-Dimensional Wave Equation

In this section, several numerical schemes for the one-dimensional wave equation will be analyzed theoretically and numerically. The numerical solutions obtained using the different schemes will then be compared for various CFL numbers. Their accuracy and stability will be investigated, and a spatial convergence check is given in Appendix G. The theoretical derivations are collected in the appendices; the main text only reports their results.

1.1 Summary of the Theoretical Analysis

A theoretical analysis was performed for three families of schemes: the hybrid explicit–implicit scheme with $\theta = \{0, \frac{1}{2}, 1\}$ (Appendices A–C), a second-order-in-space and fourth-order-in-time scheme, hereafter the (2, 4) scheme (Appendices D and E), and a fourth-order-in-space and second-order-in-time scheme, hereafter the (4, 2) scheme (Appendix F), where the notation (p, q) denotes the spatial and temporal orders, respectively. Consistency was assessed using a Taylor-series analysis of the local truncation error τ , while stability was studied using von Neumann analysis with $\nu = c\Delta t/\Delta x$. All five schemes are consistent with the advection equation. For the periodic interior problem considered in the von Neumann analysis, consistency together with stability implies convergence by the Lax–Richtmyer equivalence theorem [1]. The main theoretical results are summarized in Table 1.

Table 1: Summary of the theoretical analysis

Scheme	Leading truncation error	Order (x, t)	Stability	Convergence
Hybrid, $\theta = 0$	$\mathcal{O}(\Delta t) + \mathcal{O}(\Delta x^2)$	(2, 1)	Unstable	No
Hybrid, $\theta = \frac{1}{2}$	$\mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta x^2)$	(2, 2)	Neutral, $\forall \nu$	Yes
Hybrid, $\theta = 1$	$\mathcal{O}(\Delta t) + \mathcal{O}(\Delta x^2)$	(2, 1)	Stable, $\forall \nu$	Yes
(2, 4) scheme	$\mathcal{O}(\Delta t^4) + \mathcal{O}(\Delta x^2)$	(2, 4)	$ \nu \leq \sqrt{3}$	Yes, if $ \nu \leq \sqrt{3}$
(4, 2) scheme	$\mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta x^4)$	(4, 2)	Neutral, $\forall \nu$	Yes

1.2 Numerical Methodology

All schemes were implemented in C++17 with Eigen. The advection equation is solved on $[0, \pi]$ with $c = 0.5$ on $N = 1201$ points ($\Delta x = \pi/1200$), starting from the top-hat pulse $u_0(x) = 1$ for $0.5 \leq x \leq 1.0$ and 0 elsewhere. The exact solution is the translated pulse $u(x, t) = u_0(x - ct)$: its midpoint $x = 0.75$ reaches $x = 2.5$ at $t_f = (2.5 - 0.75)/c = 3.5$, where the exact support is $[2.25, 2.75]$. This lies below π , so the pulse never reaches the right boundary. The requested CFL values are 0.25, 0.50, 0.75, 1.00, and 1.25; Δt is reduced slightly so that t_f is reached in a whole number of steps, so the effective CFL is marginally lower (e.g. 0.9992 for 1.00 and 1.2494 for 1.25). The hybrid θ scheme and the (4, 2) scheme are implicit: their constant sparse matrices are factorized once and reused at every step. Leapfrog starts with one backward-upwind step. Endpoint values are held at their initial zero value (one point per end for the θ scheme, two for the five-point schemes), and backward upwind fixes the inlet. The final profiles are compared with the exact pulse using $E_2 = [\Delta x \sum_j (u_j - u_{\text{exact},j})^2]^{1/2}$.

1.3 Results and Discussion

1.3.1 Influence of the CFL Number

Table 2 reports the error E_2 at t_f for each scheme and requested CFL.

Table 2: Error E_2 at t_f versus requested CFL

Scheme	0.25	0.50	0.75	1.00	1.25
Backward upwind	0.166	0.150	0.126	0.032	div.
Forward upwind	div.	div.	div.	div.	div.
Lax	0.248	0.197	0.156	0.025	div.
Lax–Wendroff	0.120	0.111	0.099	0.041	div.
Leapfrog	0.138	0.132	0.121	0.053	div.
Hybrid, $\theta = 0$	div.	div.	div.	div.	div.
Hybrid, $\theta = \frac{1}{2}$	0.142	0.143	0.147	0.151	0.156
Hybrid, $\theta = 1$	0.127	0.150	0.166	0.178	0.188
(2, 4) scheme	0.121	0.117	0.114	0.112	0.110
(4, 2) scheme	0.093	0.105	0.117	0.127	0.136

The forward scheme and the hybrid scheme with $\theta = 0$ diverge at every CFL, as the von Neumann analysis predicts. Backward upwind, Lax, Lax–Wendroff, and leapfrog stay bounded up to CFL = 1 and diverge at 1.25, confirming a limit of CFL ≤ 1 . The (2, 4) scheme is still bounded at 1.25, consistent with its limit $|\nu| \leq \sqrt{3}$, and the hybrid schemes with $\theta = \frac{1}{2}$ and $\theta = 1$ and the (4, 2) scheme are bounded over the whole sweep.

Backward upwind, Lax, Lax–Wendroff, and leapfrog improve as CFL approaches 1 (Figure 1). Their error drops sharply at CFL = 1 (backward upwind: $0.150 \rightarrow 0.032$ from CFL 0.5 to 1) because the stencil then shifts the pulse by exactly one cell per step and the numerical diffusion of the first-order schemes vanishes. The implicit schemes show no such benefit: their error increases with CFL (Table 2). They remain stable at large CFL, but a larger step only adds time-integration error and damping, whereas the explicit (2, 4) scheme is nearly insensitive to CFL.

The schemes of second order or higher (Lax–Wendroff, leapfrog, the hybrid $\theta = \frac{1}{2}$ scheme, (2, 4), (4, 2)) overshoot the pulse at every stable CFL. The only exception is Lax–Wendroff at CFL = 1, where it reduces to the exact shift. This is dispersion at the two jumps, not instability, and it does not vanish as CFL decreases. Only the dissipative schemes (backward upwind, Lax, $\theta = 1$) stay within $[0, 1]$. The profiles of all bounded schemes at CFL = 0.50 are shown in Appendix H.

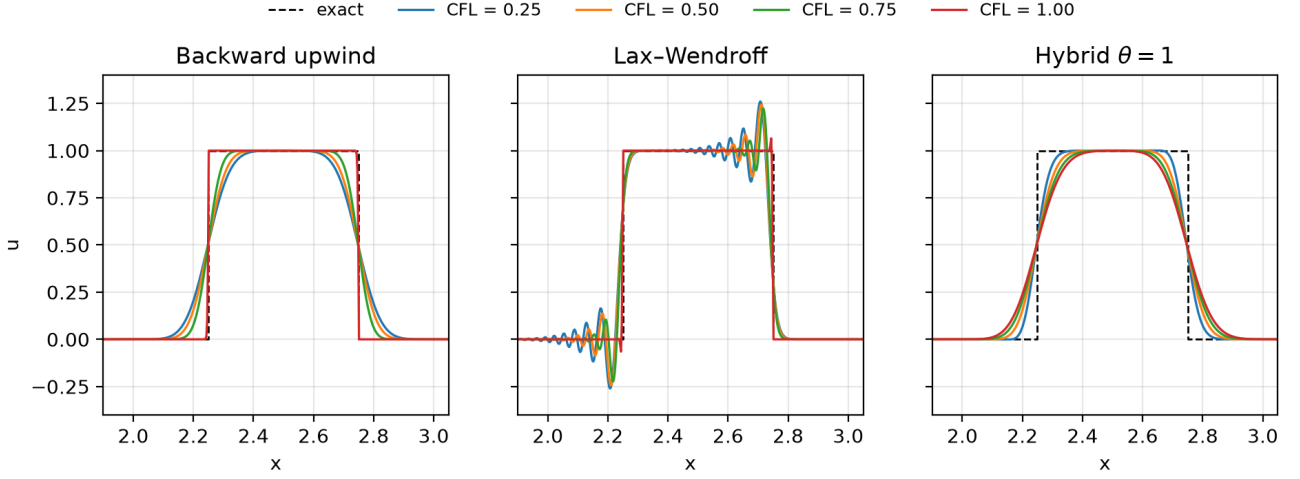


Figure 1: Profiles at t_f against the exact pulse (dashed) for CFL = 0.25, 0.50, 0.75, 1.00: backward upwind (left), Lax–Wendroff (centre), and hybrid $\theta = 1$ (right).

2 Quasi-One-Dimensional Euler Equations

We solve the quasi-one-dimensional Euler equations for compressible flow through a variable-area nozzle:

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} = S, \quad Q = \begin{pmatrix} \rho A \\ \rho u A \\ \rho E A \end{pmatrix}, \quad F = \begin{pmatrix} \rho u A \\ (\rho u^2 + p) A \\ u(\rho E + p) A \end{pmatrix}, \quad S = \begin{pmatrix} 0 \\ p dA/dx \\ 0 \end{pmatrix} \quad (1)$$

Here, x and t are position and time, $A(x)$ is the nozzle area, ρ is density, u is velocity, p is pressure, and E is total energy per unit mass. For an ideal gas, $p = (\gamma - 1)\rho(E - u^2/2)$, where γ is the ratio of specific heats.

2.1 Numerical implementation

The nozzle is discretized with 201 points over $0 \leq x \leq 10$ m, giving $\Delta x = 0.05$ m. Both test cases impose an inlet Mach number of $M_{\text{in}} = 1.25$. In the first case, the outlet is supersonic and the flow expands smoothly through the nozzle. In the second, a subsonic outlet pressure $p_b = 1.9p_{\text{in}}$ is imposed, producing a normal shock inside the nozzle. We compare the explicit MacCormack and implicit Beam–Warming [2] methods; their implementations and stopping criteria are detailed in Appendix I.

2.2 MacCormack Results

Figure 2 shows that the flow enters at $M = 1.25$ and accelerates to $M_{\text{out}} \approx 2.035$ with a supersonic outlet. This agrees with the expected expansion as the nozzle area increases. With a subsonic outlet, the flow accelerates before crossing a shock near $x = 4.9$ m, then remains subsonic downstream. Oscillations near the shock produce a peak of $M \approx 1.94$. The Mach profiles vary little over CFL = 0.25–1.

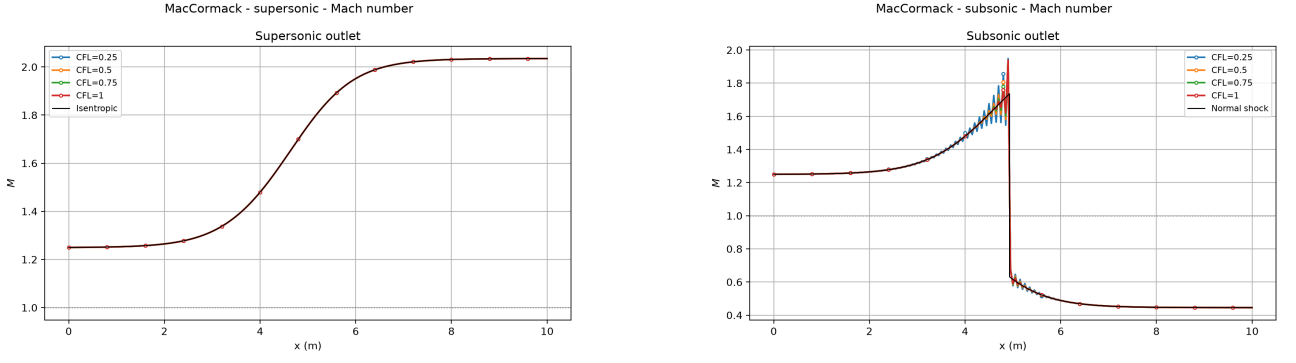


Figure 2: MacCormack Mach profiles: supersonic (left) and subsonic (right) outlets.

All four cases reach the 10^{-8} relative-update stopping threshold (Figure 3). Increasing CFL from 0.25 to 1 reduces the iteration count from about 5,300 to 1,400 supersonically and from 27,000 to 6,200 subsonically. Convergence is more oscillatory in the shock case.

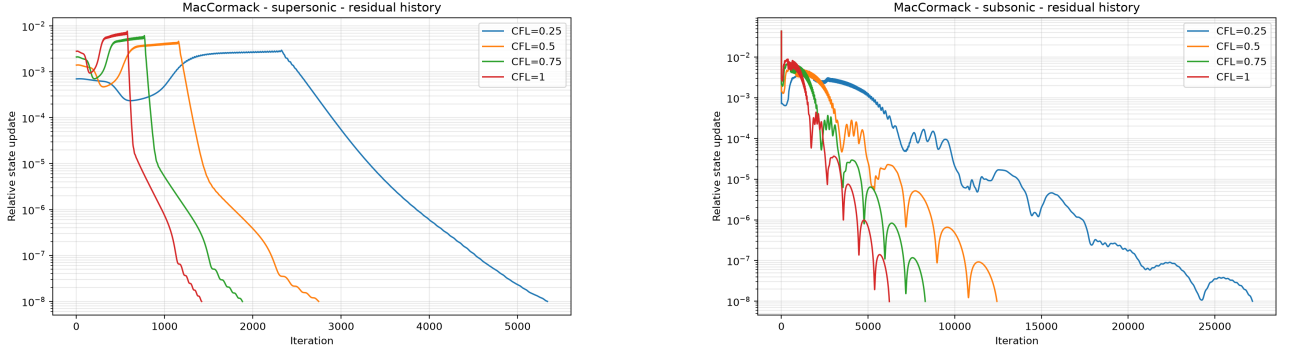


Figure 3: MacCormack relative state updates: supersonic (left) and subsonic (right) outlets.

2.3 Beam-Warming Results

Figure 4 shows continuous supersonic acceleration from the imposed inlet Mach number of 1.25, following the isentropic reference [3]. With a subsonic outlet, a shock near $x = 4.9$ m slows the flow to subsonic speed. The numerical peak reaches $M \approx 2.09$; the oscillations are concentrated near the shock. The profiles are nearly superposed for $CFL = 1$ –10.

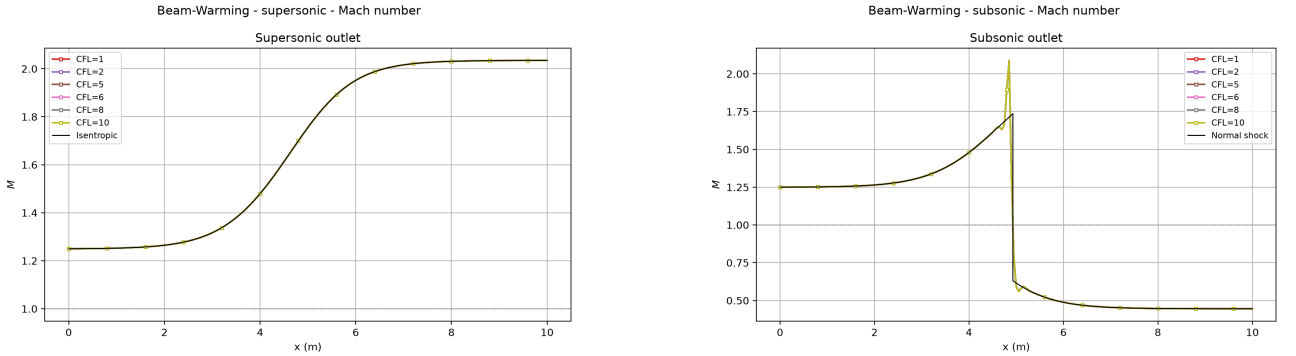


Figure 4: Beam-Warming Mach profiles: supersonic (left) and subsonic (right) outlets.

All six cases satisfy the 10^{-8} equation-residual stopping criterion. Figure 5 instead shows their relative state updates. From $CFL = 1$ to 10, the iteration count falls from about 1,240 to 150 supersonically and from 6,300 to 760 subsonically.

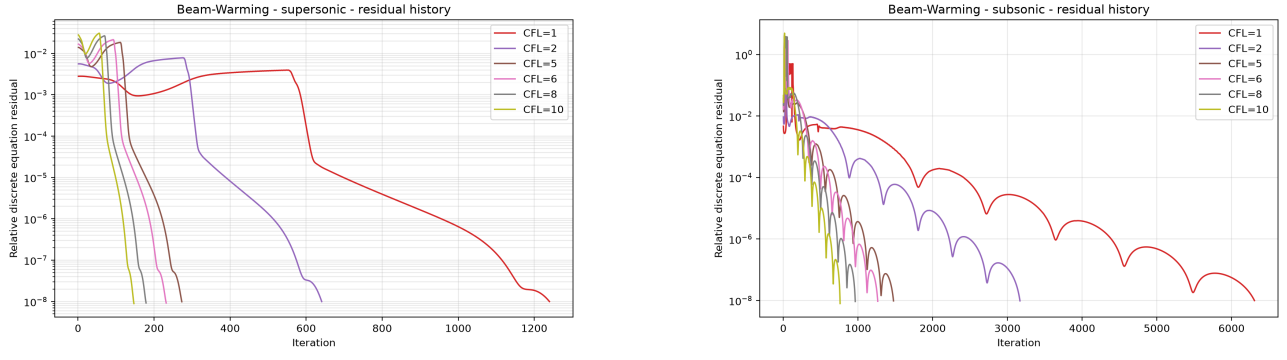


Figure 5: Beam-Warming relative state updates: supersonic (left) and subsonic (right) outlets.

2.4 Comparison

Both methods reproduce the smooth supersonic solution and locate the shock near $x = 4.9$ m. At $CFL = 1$, MacCormack has a smaller Mach peak near the shock (1.94 versus 2.09), while Beam-Warming's oscillations are more localized. The tested profiles converge up to $CFL = 1$ for MacCormack and $CFL = 10$ for Beam-Warming. Figure 6 shows that, at $CFL = 1$, MacCormack takes 3.40 versus 22.14 s subsonically and 0.78 versus 4.33 s supersonically. At $CFL = 10$, Beam-Warming takes 2.68 and 0.54 s, respectively. Each scheme uses a stopping criterion suited to its method, so these timing comparisons remain indicative.

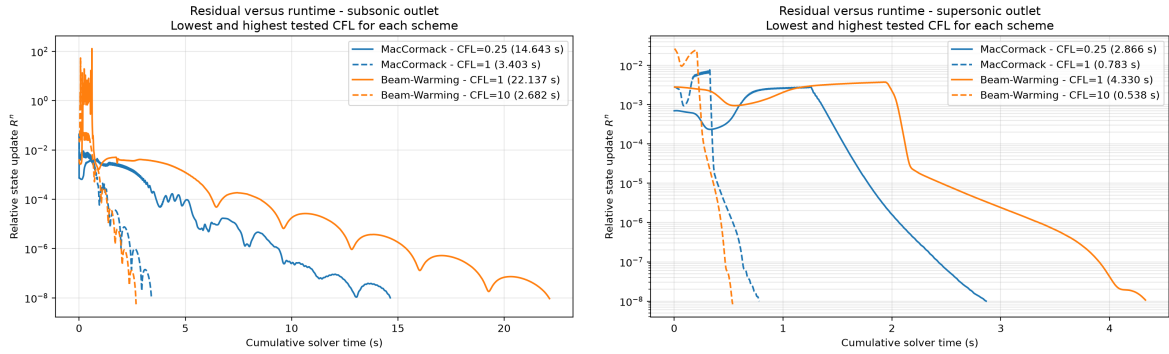


Figure 6: Relative state update versus cumulative runtime: subsonic (left) and supersonic (right).

References

- [1] P. D. Lax and R. D. Richtmyer, *Survey of the Stability of Linear Finite Difference Equations*, Communications on Pure and Applied Mathematics, 9, 267–293 (1956). Original paper.
- [2] R. M. Beam and R. F. Warming, *An Implicit Finite-Difference Algorithm for Hyperbolic Systems in Conservation-Law Form*, Journal of Computational Physics, 22, 87–110 (1976). doi:10.1016/0021-9991(76)90110-8.
- [3] Ames Research Staff, *Equations, Tables, and Charts for Compressible Flow*, NACA Report 1135 (1953). NASA report.

A Development for hybrid scheme's useful form

The hybrid explicit-implicit scheme, analyzed for $\theta = \{0, 0.5, 1\}$, is:

$$u_j^{n+1} + \theta \frac{\text{CFL}}{2} (u_{j+1}^{n+1} - u_{j-1}^{n+1}) = u_j^n - (1 - \theta) \frac{\text{CFL}}{2} (u_{j+1}^n - u_{j-1}^n) \quad (2)$$

We start from the linear advection equation:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \quad (3)$$

The CFL number can be defined as:

$$\nu \equiv \text{CFL} = \frac{c \Delta t}{\Delta x} \quad (4)$$

We can substitute 4 in the original form of the scheme (2) and obtain:

$$u_j^{n+1} + \theta \frac{\nu}{2} (u_{j+1}^{n+1} - u_{j-1}^{n+1}) = u_j^n - (1 - \theta) \frac{\nu}{2} (u_{j+1}^n - u_{j-1}^n) \quad (5)$$

We then bring everything to left and divide by Δt . Since $\frac{\nu}{2\Delta t} = \frac{c}{2\Delta x}$ we get:

$$\boxed{\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \left[\theta \frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} + (1 - \theta) \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} \right] = 0} \quad (6)$$

The centered spatial operator is:

$$D_0 u_j^n = \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} \quad (7)$$

So, the scheme can also be rewritten as:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c [\theta D_0 u_j^{n+1} + (1 - \theta) D_0 u_j^n] = 0 \quad (8)$$

B Taylor Expansions for Hybrid Explicit-Implicit Scheme

B.0.0.1 Time derivative approximation. Expand u_j^{n+1} about (x_j, t_n) :

$$u_j^{n+1} = u_j^n + \Delta t u_t + \frac{\Delta t^2}{2} u_{tt} + \frac{\Delta t^3}{6} u_{ttt} + \mathcal{O}(\Delta t^4) \quad (9)$$

Therefore:

$$\boxed{\frac{u_j^{n+1} - u_j^n}{\Delta t} = u_t + \frac{\Delta t}{2} u_{tt} + \frac{\Delta t^2}{6} u_{ttt} + \mathcal{O}(\Delta t^3)} \quad (10)$$

B.0.0.2 Centered spatial derivative at t_n . We first expand the neighboring points:

$$u_{j+1}^n = u + \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} + \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} + \dots \quad (11)$$

And :

$$u_{j-1}^n = u - \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} - \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} + \dots \quad (12)$$

Subtracting $u_{j+1}^n - u_{j-1}^n$, we get:

$$u_{j+1}^n - u_{j-1}^n = 2\Delta x u_x + \frac{\Delta x^3}{3} u_{xxx} + \mathcal{O}(\Delta x^5) \quad (13)$$

This can be rewritten as:

$$\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = u_x + \frac{\Delta x^2}{6} u_{xxx} + \mathcal{O}(\Delta x^4) \quad (14)$$

Hence:

$$\boxed{D_0 u_j^n = u_x + \frac{\Delta x^2}{6} u_{xxx} + \mathcal{O}(\Delta x^4)} \quad (15)$$

B.0.0.3 Spatial derivative at t_{n+1} . At the next time step, we get:

$$D_0 u_j^{n+1} = u_x(x_j, t_{n+1}) + \frac{\Delta x^2}{6} u_{xxx}(x_j, t_{n+1}) + \mathcal{O}(\Delta x^4) \quad (16)$$

Expanding the derivatives in time, we get:

$$u_x(x_j, t_{n+1}) = u_x + \Delta t u_{xt} + \frac{\Delta t^2}{2} u_{xtt} + \mathcal{O}(\Delta t^3) \quad (17)$$

And:

$$u_{xxx}(x_j, t_{n+1}) = u_{xxx} + \Delta t u_{xxxt} + \mathcal{O}(\Delta t^2) \quad (18)$$

Therefore, we get:

$$\boxed{D_0 u_j^{n+1} = u_x + \Delta t u_{xt} + \frac{\Delta t^2}{2} u_{xtt} + \frac{\Delta x^2}{6} u_{xxx} + \mathcal{O}(\Delta t^3, \Delta t \Delta x^2, \Delta x^4)} \quad (19)$$

B.0.0.4 Substituting back into the scheme. The local truncation error is:

$$\tau = \frac{u_j^{n+1} - u_j^n}{\Delta t} + c[\theta D_0 u_j^{n+1} + (1 - \theta) D_0 u_j^n] \quad (20)$$

Substituting the Taylor expansions gives:

$$\begin{aligned} \tau = u_t + \frac{\Delta t}{2} u_{tt} + \frac{\Delta t^2}{6} u_{ttt} + c(1 - \theta) \left(u_x + \frac{\Delta x^2}{6} u_{xxx} \right) \\ + c\theta \left(u_x + \Delta t u_{xt} + \frac{\Delta t^2}{2} u_{xtt} + \frac{\Delta x^2}{6} u_{xxx} \right) + \dots \end{aligned} \quad (21)$$

We can then combine the u_x terms as such:

$$c(1 - \theta)u_x + c\theta u_x = cu_x \quad (22)$$

Thus, we can write:

$$\tau = u_t + cu_x + \Delta t \left(\frac{1}{2} u_{tt} + c\theta u_{xt} \right) + \Delta t^2 \left(\frac{1}{6} u_{ttt} + \frac{c\theta}{2} u_{xtt} \right) + \frac{c\Delta x^2}{6} u_{xxx} + \dots \quad (23)$$

Since the exact solution satisfies $u_t + cu_x = 0$, the first two terms cancel out. We now need to express the time derivatives in terms of spatial derivatives. From:

$$u_t = -cu_x \quad (24)$$

By differentiating with respect to t , we get:

$$u_{tt} = -cu_{xt} \quad (25)$$

And with respect to x :

$$u_{xt} = -cu_{xx} \quad (26)$$

Therefore, we get:

$$u_{tt} = c^2 u_{xx} \quad (27)$$

For the next derivatives, we get:

$$u_{ttt} = -c^3 u_{xxx} \quad (28)$$

And:

$$u_{xtt} = c^2 u_{xxx} \quad (29)$$

After substituting these relations into the truncation error, we get:

$$\tau = \Delta t \left[\frac{1}{2} c^2 u_{xx} - c^2 \theta u_{xx} \right] + \Delta t^2 \left[\frac{-c^3}{6} u_{xxx} + \frac{c^3 \theta}{2} u_{xxx} \right] + \frac{c\Delta x^2}{6} u_{xxx} + \dots \quad (30)$$

Hence:

$$\boxed{\tau = c^2 \left(\frac{1}{2} - \theta \right) \Delta t u_{xx} + c^3 \left(\frac{\theta}{2} - \frac{1}{6} \right) \Delta t^2 u_{xxx} + \frac{c\Delta x^2}{6} u_{xxx} + \dots} \quad (31)$$

This is the key expression for the consistency result. From this point, we will be looking at 3 cases for θ values and determine whether the scheme is consistent with the advection equation.

B.0.0.5 Case $\theta = 0$. In this case, equation 31 becomes:

$$\tau = \frac{c^2}{2}\Delta t u_{xx} - \frac{c^3}{6}\Delta t^2 u_{xxx} + \frac{c\Delta x^2}{6}u_{xxx} + \dots \quad (32)$$

The leading errors are therefore:

$$\boxed{\tau = \mathcal{O}(\Delta t) + \mathcal{O}(\Delta x^2)} \quad (33)$$

This shows that the scheme is first order in time and second order in space. It is the well-known forward-Euler, centered-space scheme. And since:

$$\boxed{\lim_{\Delta x, \Delta t \rightarrow 0} \tau = 0} \quad (34)$$

We can conclude that this scheme is consistent with the advection equation.

B.0.0.6 Case $\theta = \frac{1}{2}$: Crank-Nicolson. In this case, equation 31 becomes:

$$\tau = \frac{c^3\Delta t^2}{12}u_{xxx} + \frac{c\Delta x^2}{6}u_{xxx} + \dots \quad (35)$$

The leading errors are:

$$\boxed{\tau = \mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta x^2)} \quad (36)$$

This is second order in both time and space. The scheme is also consistent with the advection equation, since:

$$\boxed{\lim_{\Delta x, \Delta t \rightarrow 0} \tau = 0} \quad (37)$$

B.0.0.7 Case $\theta = 1$. In this case, equation 31 becomes:

$$\tau = -\frac{c^2}{2}\Delta t u_{xx} + \frac{c^3}{3}\Delta t^2 u_{xxx} + \frac{c\Delta x^2}{6}u_{xxx} + \dots \quad (38)$$

The leading errors are:

$$\boxed{\tau = \mathcal{O}(\Delta t) + \mathcal{O}(\Delta x^2)} \quad (39)$$

It therefore is first order in time and second order in space. This value of θ gives a fully implicit scheme, which is also consistent with the advection equation since, once again:

$$\boxed{\lim_{\Delta x, \Delta t \rightarrow 0} \tau = 0} \quad (40)$$

C Von Neumann Stability Analysis for Hybrid Scheme

For a linear constant-coefficient PDE, we use a Fourier mode of the form:

$$u_j^n = \hat{u}^n e^{ij\phi} = G^n e^{ij\phi} \quad (41)$$

Where G is the amplification factor. Then, we have:

$$u_{j+1}^n = G^n e^{ij\phi} e^{i\phi} \quad (42)$$

And

$$u_{j-1}^n = G^n e^{ij\phi} e^{-i\phi} \quad (43)$$

Therefore:

$$u_{j+1}^n - u_{j-1}^n = G^n e^{ij\phi} (e^{i\phi} - e^{-i\phi}) \quad (44)$$

We can use the following property:

$$e^{i\phi} - e^{-i\phi} = 2i \sin \phi \quad (45)$$

This gives:

$$u_{j+1}^n - u_{j-1}^n = 2i \sin \phi G^n e^{ij\phi} \quad (46)$$

At time $n + 1$ we have:

$$u_j^{n+1} = G u_j^n \quad (47)$$

So similarly:

$$u_{j+1}^{n+1} - u_{j-1}^{n+1} = 2i \sin \phi G u_j^n \quad (48)$$

We can now substitute these values into the scheme and obtain:

$$G u_j^n + \theta \frac{\nu}{2} (2i \sin \phi G u_j^n) = u_j^n - (1 - \theta) \frac{\nu}{2} (2i \sin \phi u_j^n) \quad (49)$$

By dividing by u_j^n and isolating G , we get:

$$\boxed{G = \frac{1 - i(1 - \theta)\nu \sin \phi}{1 + i\theta\nu \sin \phi}} \quad (50)$$

Von Neumann stability requires $|G| \leq 1$. By taking the magnitude squared, we get:

$$|G|^2 = \frac{1 + (1 - \theta)^2 \nu^2 \sin^2 \phi}{1 + \theta^2 \nu^2 \sin^2 \phi} \quad (51)$$

We then obtain the following inequality:

$$(1 - \theta)^2 \nu^2 \sin^2 \phi \leq \theta^2 \nu^2 \sin^2 \phi \quad (52)$$

We simplify to get:

$$(1 - \theta)^2 \leq \theta^2 \quad (53)$$

By expanding and simplifying further we get:

$$1 - 2\theta \leq 0 \quad (54)$$

And therefore:

$$\boxed{\theta \geq \frac{1}{2}} \quad (55)$$

C.0.0.1 Case-by-case conclusions. For $\theta = 0$, $|G| > 1$ whenever $\nu \sin \phi \neq 0$. At $\theta = 1/2$ (Crank–Nicolson time integration), $|G| = 1$: modal amplitudes are preserved and only their phase changes. At $\theta = 1$, $|G| < 1$ for $\nu \sin \phi \neq 0$, so the implicit scheme damps those modes. The zero and Nyquist modes remain neutral in all cases.

C.0.0.2 Convergence. For a well-posed linear periodic problem, consistency together with stability gives convergence by the Lax–Richtmyer equivalence theorem [1]. The three cases are summarized below.

Table 3: Convergence of the hybrid explicit-implicit scheme.

θ	Consistency	Stability	Convergence
0	Consistent	Unstable	Not convergent
0.5	Consistent	Neutral for all CFL	Convergent
1	Consistent	Damping for all CFL	Convergent

D Derivation of the 2nd Order Space and 4th Order Time Scheme

To achieve fourth-order temporal accuracy, the solution at time t^{n+1} is expanded about t^n using a fourth-order Taylor expansion:

$$u_j^{n+1} = u_j^n + \Delta t \left. \frac{\partial u}{\partial t} \right|_j^n + \frac{\Delta t^2}{2} \left. \frac{\partial^2 u}{\partial t^2} \right|_j^n + \frac{\Delta t^3}{6} \left. \frac{\partial^3 u}{\partial t^3} \right|_j^n + \frac{\Delta t^4}{24} \left. \frac{\partial^4 u}{\partial t^4} \right|_j^n + O(\Delta t^5) \quad (56)$$

The time derivatives appearing in Eq. (56) can be converted into spatial derivatives using the governing equation. We have:

$$u_t = -cu_x \quad (57)$$

Differentiating once with respect to time gives:

$$u_{tt} = -cu_{xt} \quad (58)$$

Differentiating the governing equation with respect to x gives:

$$u_{xt} = -cu_{xx} \quad (59)$$

And therefore:

$$u_{tt} = c^2 u_{xx} \quad (60)$$

Repeating the same procedure gives:

$$u_{ttt} = -c^3 u_{xxx} \quad (61)$$

And:

$$u_{tttt} = c^4 u_{xxxx} \quad (62)$$

Thus, in general:

$$\frac{\partial^m u}{\partial t^m} = (-c)^m \frac{\partial^m u}{\partial x^m} \quad (63)$$

Substitution into Eq. (56) yields:

$$u_j^{n+1} = u_j^n - c\Delta t u_x + \frac{c^2 \Delta t^2}{2} u_{xx} - \frac{c^3 \Delta t^3}{6} u_{xxx} + \frac{c^4 \Delta t^4}{24} u_{xxxx} + O(\Delta t^5) \quad (64)$$

where all spatial derivatives are evaluated at (x_j, t^n) .

D.1 Second-Order Spatial Approximations

Second-order centered finite-difference approximations are now constructed for each spatial derivative appearing in Eq. (64). The Taylor expansions of the neighboring points u_{j+1}^n and u_{j-1}^n about x_j are:

$$\begin{aligned} u_{j+1}^n &= u_j^n + \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} + \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} \\ &\quad + \frac{\Delta x^5}{120} u_{xxxxx} + \frac{\Delta x^6}{720} u_{xxxxxx} + O(\Delta x^7) \end{aligned} \quad (65)$$

And:

$$\begin{aligned}
u_{j-1}^n &= u_j^n - \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} - \frac{\Delta x^3}{6} u_{xxx} + \frac{\Delta x^4}{24} u_{xxxx} \\
&\quad - \frac{\Delta x^5}{120} u_{xxxxx} + \frac{\Delta x^6}{720} u_{xxxxxx} + O(\Delta x^7)
\end{aligned} \tag{66}$$

The expansions at two grid points away are similarly:

$$\begin{aligned}
u_{j+2}^n &= u_j^n + 2\Delta x u_x + 2\Delta x^2 u_{xx} + \frac{4}{3}\Delta x^3 u_{xxx} + \frac{2}{3}\Delta x^4 u_{xxxx} \\
&\quad + \frac{4}{15}\Delta x^5 u_{xxxxx} + \frac{4}{45}\Delta x^6 u_{xxxxxx} + O(\Delta x^7)
\end{aligned} \tag{67}$$

And:

$$\begin{aligned}
u_{j-2}^n &= u_j^n - 2\Delta x u_x + 2\Delta x^2 u_{xx} - \frac{4}{3}\Delta x^3 u_{xxx} + \frac{2}{3}\Delta x^4 u_{xxxx} \\
&\quad - \frac{4}{15}\Delta x^5 u_{xxxxx} + \frac{4}{45}\Delta x^6 u_{xxxxxx} + O(\Delta x^7)
\end{aligned} \tag{68}$$

D.1.1 First Derivative

Subtracting Eq. (66) from Eq. (65) gives:

$$u_{j+1}^n - u_{j-1}^n = 2\Delta x u_x + \frac{\Delta x^3}{3} u_{xxx} + O(\Delta x^5) \tag{69}$$

Dividing by $2\Delta x$ gives:

$$\boxed{u_x = \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} + O(\Delta x^2)} \tag{70}$$

Thus, the centered approximation of the first derivative is second-order accurate in space.

D.1.2 Second Derivative

Adding Eqs. (65) and (66) and subtracting $2u_j^n$ gives:

$$u_{j+1}^n - 2u_j^n + u_{j-1}^n = \Delta x^2 u_{xx} + \frac{\Delta x^4}{12} u_{xxxx} + O(\Delta x^6) \tag{71}$$

Therefore:

$$\boxed{u_{xx} = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} + O(\Delta x^2)} \tag{72}$$

D.1.3 Third Derivative

A centered approximation of the third derivative requires a five-point stencil. Consider the following combination:

$$u_{j+2}^n - 2u_{j+1}^n + 2u_{j-1}^n - u_{j-2}^n \tag{73}$$

Substituting Eqs. (65)–(68) into this expression gives:

$$u_{j+2}^n - 2u_{j+1}^n + 2u_{j-1}^n - u_{j-2}^n = 2\Delta x^3 u_{xxx} + \frac{\Delta x^5}{2} u_{xxxxx} + O(\Delta x^7) \tag{74}$$

Dividing by $2\Delta x^3$ gives:

$$\boxed{u_{xxx} = \frac{u_{j+2}^n - 2u_{j+1}^n + 2u_{j-1}^n - u_{j-2}^n}{2\Delta x^3} + O(\Delta x^2)} \quad (75)$$

D.1.4 Fourth Derivative

For the fourth derivative, consider the symmetric five-point combination:

$$u_{j-2}^n - 4u_{j-1}^n + 6u_j^n - 4u_{j+1}^n + u_{j+2}^n \quad (76)$$

Substitution of the Taylor expansions gives:

$$\begin{aligned} u_{j-2}^n - 4u_{j-1}^n + 6u_j^n - 4u_{j+1}^n + u_{j+2}^n \\ = \Delta x^4 u_{xxxx} + \frac{\Delta x^6}{6} u_{xxxxx} + O(\Delta x^8) \end{aligned} \quad (77)$$

Hence:

$$\boxed{u_{xxxx} = \frac{u_{j-2}^n - 4u_{j-1}^n + 6u_j^n - 4u_{j+1}^n + u_{j+2}^n}{\Delta x^4} + O(\Delta x^2)} \quad (78)$$

All spatial derivatives in the temporal Taylor expansion are therefore approximated with second-order spatial accuracy.

D.2 Construction of the Finite-Difference Scheme

Substituting Eqs. (70), (72), (75), and (78) into Eq. (64) gives:

$$\begin{aligned} u_j^{n+1} = u_j^n &- \frac{c\Delta t}{2\Delta x} (u_{j+1}^n - u_{j-1}^n) \\ &+ \frac{c^2\Delta t^2}{2\Delta x^2} (u_{j+1}^n - 2u_j^n + u_{j-1}^n) \\ &- \frac{c^3\Delta t^3}{12\Delta x^3} (u_{j+2}^n - 2u_{j+1}^n + 2u_{j-1}^n - u_{j-2}^n) \\ &+ \frac{c^4\Delta t^4}{24\Delta x^4} (u_{j-2}^n - 4u_{j-1}^n + 6u_j^n - 4u_{j+1}^n + u_{j+2}^n). \end{aligned} \quad (79)$$

Introducing the CFL number:

$$\text{CFL} = \frac{c\Delta t}{\Delta x}, \quad (80)$$

The final scheme becomes:

$$\boxed{\begin{aligned} u_j^{n+1} = u_j^n &- \frac{\text{CFL}}{2} (u_{j+1}^n - u_{j-1}^n) \\ &+ \frac{\text{CFL}^2}{2} (u_{j+1}^n - 2u_j^n + u_{j-1}^n) \\ &- \frac{\text{CFL}^3}{12} (u_{j+2}^n - 2u_{j+1}^n + 2u_{j-1}^n - u_{j-2}^n) \\ &+ \frac{\text{CFL}^4}{24} (u_{j-2}^n - 4u_{j-1}^n + 6u_j^n - 4u_{j+1}^n + u_{j+2}^n). \end{aligned}} \quad (81)$$

This is the finite-difference scheme used in the numerical implementation. Its truncation error is $\mathcal{O}(\Delta x^2) + \mathcal{O}(\Delta t^4)$, so it is consistent with the advection equation for smooth solutions.

E Von Neumann Stability Analysis for 2nd Order Space and 4th Order Time Scheme

Assume a Fourier mode:

$$u_j^n = G^n e^{ij\phi} \quad (82)$$

Where G is the amplification factor and $\phi = k\Delta x$ is the non-dimensional wave number. We have:

$$u_{j+r}^n = u_j^n e^{ir\phi} \quad (83)$$

And:

$$u_j^{n+1} = G u_j^n \quad (84)$$

Let $\nu = \text{CFL}$ and $s = \sin^2(\phi/2)$, so that $1 - \cos \phi = 2s$. The four stencils of Eq. (81) have the following Fourier symbols:

$$u_{j+1} - u_{j-1} \longrightarrow 2i \sin \phi, \quad (85)$$

$$u_{j+1} - 2u_j + u_{j-1} \longrightarrow 2 \cos \phi - 2 = -4s, \quad (86)$$

$$u_{j+2} - 2u_{j+1} + 2u_{j-1} - u_{j-2} \longrightarrow 2i \sin 2\phi - 4i \sin \phi = -8is \sin \phi, \quad (87)$$

$$u_{j-2} - 4u_{j-1} + 6u_j - 4u_{j+1} + u_{j+2} \longrightarrow 6 - 8 \cos \phi + 2 \cos 2\phi = 16s^2. \quad (88)$$

Substituting into the scheme and dividing by u_j^n gives:

$$G = 1 - i\nu \sin \phi - 2\nu^2 s + \frac{2}{3}i\nu^3 s \sin \phi + \frac{2}{3}\nu^4 s^2, \quad (89)$$

that is,

$$\boxed{G = 1 - 2\nu^2 s + \frac{2}{3}\nu^4 s^2 - i\nu \sin \phi \left(1 - \frac{2}{3}\nu^2 s\right)} \quad (90)$$

Using $\sin^2 \phi = 4s(1-s)$ and writing $m = \nu^2$, a direct expansion of $|G|^2$ gives:

$$|G|^2 - 1 = \frac{4m}{9} s^2 g(s), \quad g(s) = -9 + (12m - 2m^2)s + (m^3 - 4m^2)s^2. \quad (91)$$

Stability requires $|G| \leq 1$, i.e. $g(s) \leq 0$ for every $s \in [0, 1]$.

- At $s = 0$, $g(0) = -9 < 0$.
- At $s = 1$ (the Nyquist mode $\phi = \pi$), $g(1) = m^3 - 6m^2 + 12m - 9 = (m-3)(m^2 - 3m + 3)$. The second factor is always positive, so $g(1) \leq 0$ if and only if $m \leq 3$.
- For $m \leq 3 < 4$, g is a concave quadratic in s . If its vertex lies inside $[0, 1]$, then $g(s^*) = \frac{m(m-3)}{4-m} \leq 0$ for $m \leq 3$.

Hence $g(s) \leq 0$ on $[0, 1]$ if and only if $m = \nu^2 \leq 3$:

$$\boxed{|\nu| \leq \sqrt{3}} \quad (92)$$

Numerically, $\max_\phi |G| = 1$ for $\nu \leq \sqrt{3}$, while it reaches 1.056 at $\nu = 1.74$ and 1.518 at $\nu = 1.8$. Together with the consistency result of Appendix D, the Lax–Richtmyer theorem [1] gives convergence of the (2, 4) scheme for the constant-coefficient periodic problem when $|\nu| \leq \sqrt{3}$. Boundary closures require a separate stability check.

F 4th Order Space and 2nd Order Time Scheme: Consistency and Stability

The (4, 2) scheme applies the Crank–Nicolson time integration (trapezoidal rule) to the fourth-order centered derivative

$$D_4 u_j^n = \frac{u_{j-2}^n - 8u_{j-1}^n + 8u_{j+1}^n - u_{j+2}^n}{12\Delta x}, \quad \nu = \frac{c\Delta t}{\Delta x}, \quad (93)$$

that is, $\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{c}{2} (D_4 u_j^{n+1} + D_4 u_j^n) = 0$. For $j = 2, \dots, N-3$, the update is

$$u_j^{n+1} + \frac{\nu}{24} (u_{j-2}^{n+1} - 8u_{j-1}^{n+1} + 8u_{j+1}^{n+1} - u_{j+2}^{n+1}) = u_j^n - \frac{\nu}{24} (u_{j-2}^n - 8u_{j-1}^n + 8u_{j+1}^n - u_{j+2}^n). \quad (94)$$

F.0.0.1 Consistency. From the expansions of Appendix D,

$$8(u_{j+1} - u_{j-1}) - (u_{j+2} - u_{j-2}) = 12\Delta x u_x - \frac{2}{5}\Delta x^5 u_{xxxxx} + \mathcal{O}(\Delta x^7), \quad (95)$$

so that

$$D_4 u = u_x - \frac{\Delta x^4}{30} u_{xxxxx} + \mathcal{O}(\Delta x^6). \quad (96)$$

The time integration is treated as in Appendix B with $\theta = \frac{1}{2}$: the $\mathcal{O}(\Delta t)$ terms cancel, since $\frac{1}{2}u_{tt} + \frac{c}{2}u_{xt} = \frac{1}{2}(c^2 - c^2)u_{xx} = 0$, and the $\mathcal{O}(\Delta t^2)$ terms give $\left(-\frac{c^3}{6} + \frac{c^3}{4}\right)u_{xxx} = \frac{c^3}{12}u_{xxx}$. The local truncation error is therefore

$$\tau = \frac{c^3 \Delta t^2}{12} u_{xxx} - \frac{c \Delta x^4}{30} u_{xxxxx} + \mathcal{O}(\Delta t^3 + \Delta t \Delta x^4 + \Delta x^6) \quad (97)$$

so the scheme is consistent, second order in time and fourth order in space.

F.0.0.2 Stability. For a Fourier mode $u_j^n = G^n e^{ij\phi}$, the stencil of D_4 has the symbol

$$e^{-2i\phi} - 8e^{-i\phi} + 8e^{i\phi} - e^{2i\phi} = 16i \sin \phi - 2i \sin 2\phi = 2i (8 \sin \phi - \sin 2\phi), \quad (98)$$

so Eq. (94) gives $G(1 + i\mu) = 1 - i\mu$ with

$$G = \frac{1 - i\mu}{1 + i\mu}, \quad \mu = \frac{\nu}{12} (8 \sin \phi - \sin 2\phi), \quad |G| = 1, \quad (99)$$

because μ is real. Thus the periodic interior scheme is neutrally stable at any CFL and has no Fourier-mode damping; phase errors remain. Consistency and stability imply convergence for the periodic linear problem. The implementation holds two points at each boundary fixed, so its bounded-domain stability is assessed separately from this periodic Fourier result.

G Spatial Convergence Verification

Table 4: Observed spatial order $p = \ln(E_{2,i-1}/E_{2,i})/\ln(\Delta x_{i-1}/\Delta x_i)$ between successive meshes

N	B	Lax	LW	LF	$\theta=0$	$\theta=\frac{1}{2}$	$\theta=1$	(2, 4)	(4, 2)
401	0.36	0.05	1.85	1.85	1.83	1.85	1.84	1.85	3.96
801	0.51	0.10	1.99	1.99	1.87	1.99	1.87	1.99	3.95
1601	0.67	0.14	2.00	2.00	1.12	2.00	1.11	2.00	3.47
3201	0.81	0.36	2.00	2.00	div.	2.00	0.19	2.00	1.34
6401	0.95	0.79	2.01	2.01	div.	1.99	0.01	2.00	0.14
Nominal	1	1	2	2	2	2	2	2	4

H Additional Wave Equation Profiles

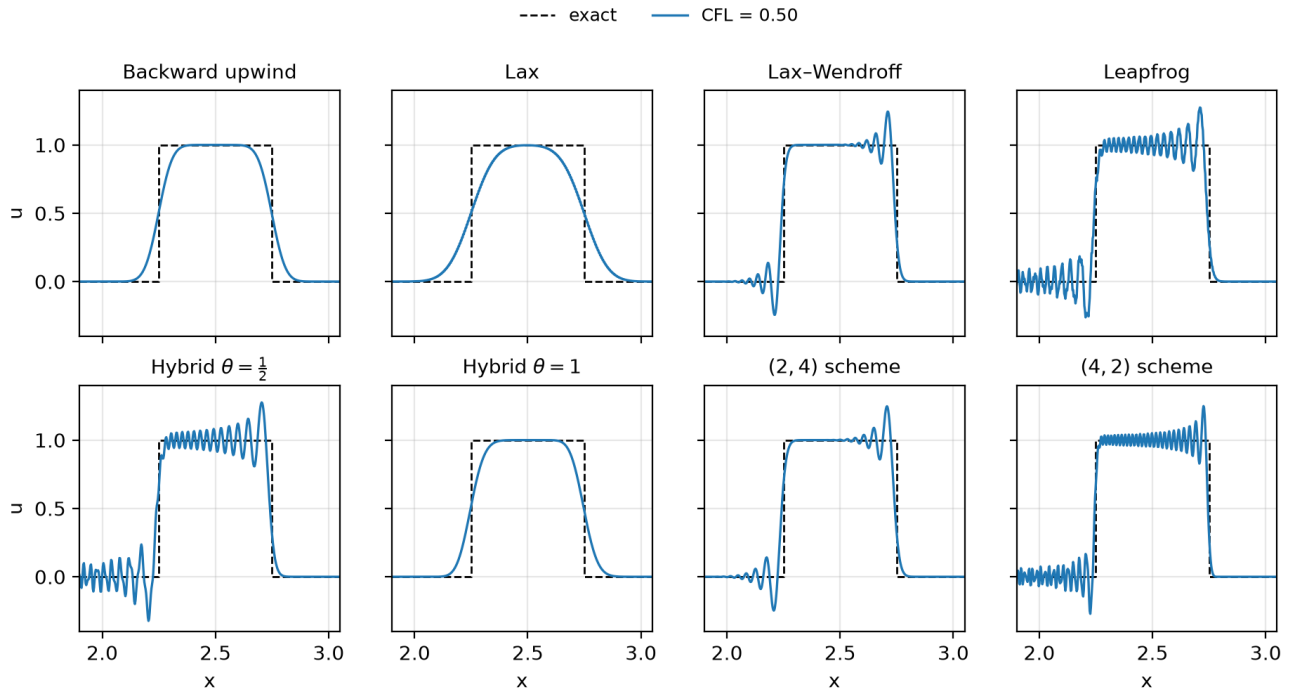


Figure 7: Profiles at t_f for $CFL = 0.50$ against the exact pulse for all bounded schemes

I Quasi-One-Dimensional Euler Solvers: Implementation Details

I.0.0.1 Setup. The nozzle area is $A(x) = 1.398 + 0.347 \tanh(0.8x - 4)$, with $dA/dx = 0.2776 \operatorname{sech}^2(0.8x - 4)$, discretized with nodes $i = 0, \dots, N$ ($N = 200$) on $0 \leq x \leq 10$ m. The gas is ideal with $\gamma = 1.4$ and $R = 287$ J/(kg K). The inlet static state is $T_{\text{in}} = 300$ K and $p_{\text{in}} = 101\,325$ Pa with $M_{\text{in}} = 1.25$, which fixes $\rho_{\text{in}} = p_{\text{in}}/(RT_{\text{in}})$, $u_{\text{in}} = M_{\text{in}}\sqrt{\gamma RT_{\text{in}}}$, and $E_{\text{in}} = RT_{\text{in}}/(\gamma - 1) + u_{\text{in}}^2/2$. Every node is initialized with the inlet state, $Q_i = (\rho_{\text{in}}A_i, \rho_{\text{in}}u_{\text{in}}A_i, \rho_{\text{in}}E_{\text{in}}A_i)^T$, and both solvers march in time to steady state with a single global time step $\Delta t = \text{CFL } \Delta x / \max_i (|u_i| + a_i)$ recomputed at every iteration, where a is the speed of sound.

I.0.0.2 Boundary conditions.

- Inlet (supersonic): the three components of Q_0 are held at the inlet state.
- Supersonic outlet: zeroth-order extrapolation, $Q_N = Q_{N-1}A_N/A_{N-1}$.
- Subsonic outlet: the static pressure $p_L = 1.9 p_{\text{in}}$ is imposed. The exit density follows from an isentropic relation with the interior node $N - 1$, $\rho_L = \rho_{N-1} (p_L/p_{N-1})^{1/\gamma}$, and the exit velocity from the outgoing Riemann invariant, $u_L = u_{N-1} + \frac{2}{\gamma-1} (a_{N-1} - a_L)$ with $a_L = \sqrt{\gamma p_L/\rho_L}$. The exit total energy is $E_L = p_L/((\gamma - 1)\rho_L) + u_L^2/2$.

I.0.0.3 MacCormack. For the interior nodes $i = 1, \dots, N - 1$,

$$Q_i^* = Q_i^n - \frac{\Delta t}{\Delta x} (F_i^n - F_{i-1}^n) + \Delta t S_i^n, \quad (100)$$

$$Q_i^{n+1} = \frac{1}{2} \left[Q_i^n + Q_i^* - \frac{\Delta t}{\Delta x} (F_{i+1}^* - F_i^*) + \Delta t S_i^* \right], \quad (101)$$

with the boundary conditions applied after each stage. No artificial dissipation is added. The iterations stop when the relative state update $R^n = \|Q^{n+1} - Q^n\|_2 / \|Q^n\|_2$ falls below 10^{-8} (cap of 30 000 iterations).

I.0.0.4 Beam-Warming. The fully implicit ($\theta = 1$) delta form is solved for $\Delta Q = Q^{n+1} - Q^n$:

$$\Delta Q_i + \frac{\Delta t}{2\Delta x} (J_{i+1}\Delta Q_{i+1} - J_{i-1}\Delta Q_{i-1}) - \sigma_i (\Delta Q_{i+1} - 2\Delta Q_i + \Delta Q_{i-1}) = \text{RHS}_i, \quad (102)$$

$$\text{RHS}_i = -\Delta t \left[\frac{F_{i+1} - F_{i-1}}{2\Delta x} - S_i \right] - \sigma_e (Q_{i-2} - 4Q_{i-1} + 6Q_i - 4Q_{i+1} + Q_{i+2}), \quad (103)$$

where $J = \partial F / \partial Q$ is the flux Jacobian,

$$J = \begin{pmatrix} 0 & 1 & 0 \\ \frac{\gamma-3}{2}u^2 & (3-\gamma)u & \gamma-1 \\ u(\frac{\gamma-1}{2}u^2 - H) & H - (\gamma-1)u^2 & \gamma u \end{pmatrix}, \quad H = \frac{\rho E + p}{\rho}. \quad (104)$$

The source term is treated explicitly. Fourth-order explicit and second-order implicit dissipation are added, with $\sigma_e = \varepsilon_e \Delta t \lambda_{\text{max}} / \Delta x$ and $\sigma_i = \varepsilon_i \Delta t \lambda_{\text{max}} / \Delta x$, where $\lambda_{\text{max}} = \max_i (|u_i| + a_i)$, $\varepsilon_e = 0.085$ ($< \frac{1}{8}$), and $\varepsilon_i = 0.2$ ($> 2\varepsilon_e$), following the course notes. The fourth-difference term is applied for $2 \leq i \leq N - 2$ only. The block-tridiagonal system of 3×3 blocks is solved with the block Thomas algorithm, with $\Delta Q = 0$ at the two boundary nodes, which are then reset

by the boundary conditions. The update is $Q^{n+1} = Q^n + \omega \Delta Q$ with $\omega = 1$, halved (up to 60 times) until the density and pressure are positive at every node. The iterations stop when the relative equation residual $\|\text{RHS}\|_2/\|Q\|_2$, evaluated before the update, falls below 10^{-8} ; the plotted histories use the relative state update R^n for both solvers.

I.0.0.5 Reference solutions. The “isentropic” curves are the supersonic branch of the isentropic area–Mach relation [3],

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}, \quad (105)$$

with A^* fixed by the inlet Mach number. The “normal shock” curves join this branch to a second isentropic branch across a normal shock, whose position is found by bisection so that the exit static pressure equals $1.9 p_{\text{in}}$.